

Predictive Analytics - Exam

Housing prices in København S and surrounding neighbourhoods

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Tabel 1: Stationarity Test Results

Area	Kbh_K	Kbh_V	Frb_C	Frb	Kbh_Ø	Kbh_N	Kbh_S
ADF_stat	-7.735	-7.395	-6.344	-5.501	-6.140	-6.809	-4.545
KPSS_stat	0.156	0.241	0.105	0.128	0.143	0.130	0.154
Stationary	Yes	Yes	Yes	Yes	Yes	Yes	Yes

1 Introduction

The housing market in Copenhagen has experienced significant price increases and variation in the city's neighbourhoods over the past decades. Understanding the development in prices of apartments can be very useful for future and present homeowners, as well as policy makers and investors. The neighbourhoods in Copenhagen each have their own characteristics, and have developed at different rates and points in time. This paper will focus on analysing the average quarterly square meter prices of owner-occupied apartments in København S (Kbh_S) from 1992 to today, and forecasting the prices 5 years ahead. Since price development in one part of the city is likely to effect others, the analysis also explores whether prices in other neighbourhoods can contribute to improving the forecasts of prices in København S.

2 Data

The time series data for this paper is from the site Statistikbanken, which is published and maintained by Realkreditrådet under Finans Danmark. It contains the quarterly average square meter prices of apartments sold in the given area from 1992 Q1 to 2024 Q4, resulting in 132 observations for each area. The neighbourhoods are defined by their postal codes. In this paper we include the neighbourhoods København K, København V, Frederiksberg C, Frederiksberg, København Ø, København N and København S.

Firstly the prices from Kbh S are investigated. The prices are plotted in R and shown below. In figure 1 we see a clear increasing trend in the prices which aligns with the ACF plot as well, where all lags show significant autocorrelation. However there's is also a large dip around 2008-2010 where the prices drop significantly, before resuming their increasing trajectory. This shows the impact that the financial crisis had on the housing prices. Additionally, the amplitude of the fluctuations appear to grow after 2005, suggesting increasing variance.

First differencing the raw price data removed the trend, but left significant heteroscedasticity. To stabilise this variance, a log transformation is applied prior to the differencing.

To assess the stationarity of the log differenced prices, both a KPSS and an ADF test are performed. The tests are done for all neighbourhoods, to ensure all data used in analysis and forecasting is stabilised.

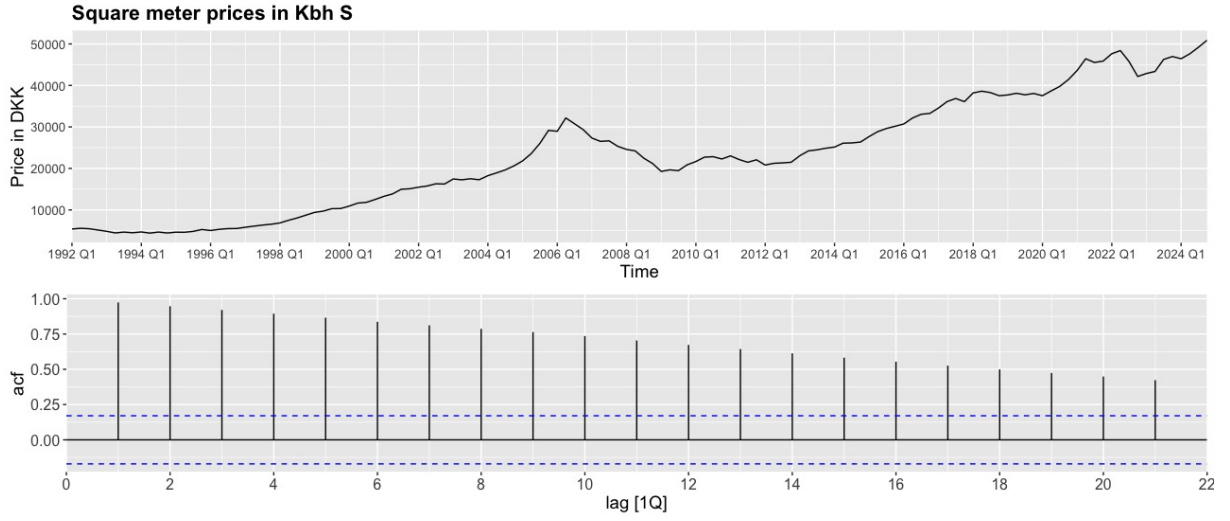


Figure 1: Plot and ACF plot of square meter prices in Kbh S.

Predictor area	Kbh Ø	Frb	Kbh N	Frb C	Frb C	Kbh K
lag	1	1	1	2	1	1
cc	0.512	0.505	0.463	0.381	0.373	0.333

Tabel 2: Top 6 Correlations

In the ADF test, a drift term was included, since the series may fluctuate around a non-zero mean, but is not expected to follow a trend. The KPSS test was done with the specification "mu", for this same reason. Both test were used, as they assess the hypothesis from opposite perspectives. The test results are summarised in table 1. With a KPSS 10% critical value of 0.347 and an ADF 1% critical value of -3.46 both tests indicate that the log differenced price series is stationary for all neighbourhoods.

2.1 Neighbourhood Correlations

To assess whether the other neighbourhoods might be helpful in predicting the prices in Kbh S the correlations between Kbh S and different lags of the other neighbourhoods are calculated. To ensure stationarity all series are the log differenced prices. Since we want to be able to predict future prices with the known information from the other neighbourhoods, it would be most interesting to look at the correlations of actual lagged values. Therefore a threshold of $lag < 0$ is implemented, as well as $correlation \geq 0.3$ to ensure a certain level of correlation. Of the values that pass these criteria, the top 6 most correlated are selected, to avoid extremely large models. The results are shown in table 2 and shows that Kbh Ø, Frb, and Kbh N have the highest correlation, this makes sense, as these are some of the most "trendy" areas in Copenhagen, with Kbh N gaining a lot of popularity in the later years, which is also the case with Kbh S.

3 Methodology

To model and forecast the dynamics of the apartment prices in Kbh S, both an ARIMA, including ARIMAX and a VAR model is and tested.

The ARIMA model is well suited for this because it captures the temporal dependence and autoregressive structure in a single variable's history. Log differencing the series makes it stationary with stable variance and limited autocorrelation. This fits well with low-order ARIMA modelling. Furthermore, ARIMA models are very simple to estimate and interpret, and performs well in the absence of strong seasonality, which appear to be limited in this data.

To address the influence of the other neighbourhoods an ARIMAX model is estimated by incorporating the lagged log returns from other areas as exogenous regressors. The lags and neighbourhoods are selected based on the correlations in section 2.1. ARIMAX keeps the simplicity from ARIMA, while letting the model use information from other areas, to hopefully improve the forecast. This is very useful when the series is influenced by leading factors, however, this influence is assumed to be one-way in the ARIMAX model (other areas influence on Kbh S)

To model possible mutual feedback between neighbourhoods, where the development in Kbh S might also affect the other areas, a VAR model is used. the VAR model allows for these feedback relationships and treats each variable as endogenous. VAR makes it easier to understand system-wide behaviour, with the cost of higher parameter burden and less clear separation between predictors and outcomes.

By comparing ARIMAX and VAR it is possible to see whether exogeneity or mutual dynamics better capture the relationships between neighbourhoods in the housing market. To evaluate model performance in a realistic forecasting setting the data is split into a training set and a test set. The training set contains all observations from 1992 Q1 to 2019 Q4 resulting in a total of 112 observations, while the test set contains the remaining 20 observations from 2020 Q1 to 2024 Q4. Models are fitted using the training set and forecasting accuracy is evaluated using the test set by comparing RMSE, MAE, MAPE, and residual autocorrelation.

4 ARIMA and ARIMAX

As addressed in the preliminary data analysis it is necessary to difference the data to make it stationary. Rather than differencing the data manually, the ARIMA models will be specified with $d = 1$, which ensures that the model handles non-stationarity internally. This simplifies the process, as it handles differencing during estimation and automatically reintegrates when forecasting.

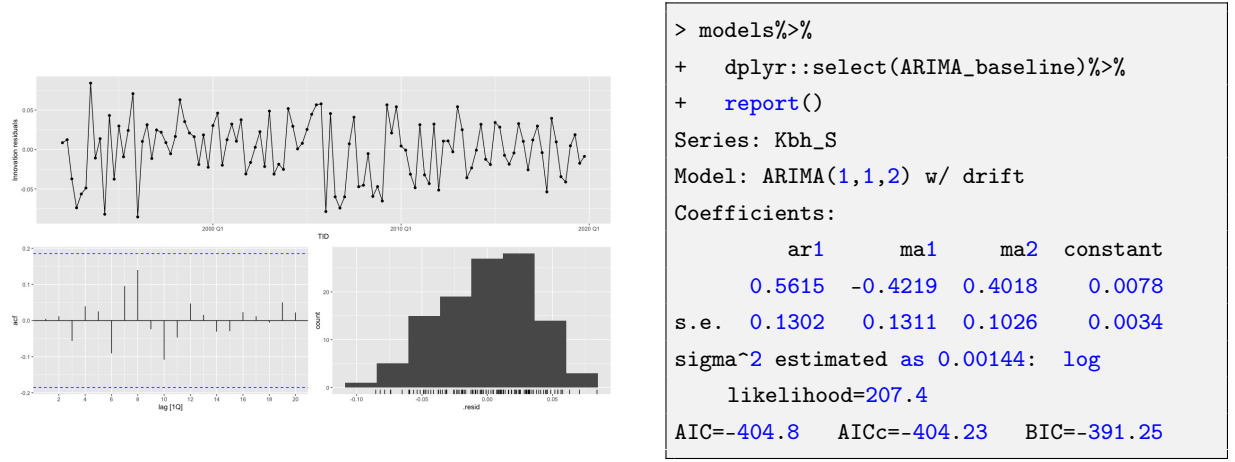


Figure 2: Residual plots and model report for ARIMA baseline model

4.1 Baseline

To start an ARIMA model will be fitted to the Kbh S log prices as a baseline. The chosen model is an ARIMA(1,1,2) with drift. This structure implies that the first differenced log prices follow an ARMA model with one AR term and two MA terms. The structure can be written as

$$\Delta \log(Kbh_S) = 0.0078 + 0.5615 \Delta \log(Kbh_S)_{t-1} - 0.4219 \cdot \varepsilon_{t-1} + 0.4018 \cdot \varepsilon_{t-2} \quad (1)$$

where the drift term (0.0078) reflects an average quarterly growth of 0.78% in the log prices. Residual diagnostics in figure 2 shows no significant autocorrelation in the residuals. It seems that the residuals behave approximately like white noise and that the model fits reasonably well.

4.2 Exogenous Variables

Now an ARIMAX model that includes the lagged log prices of other neighbourhoods, based on our correlations in table, 2 as exogenous predictors is fitted. To balance model complexity and predictive accuracy six ARIMAX models are fitted, first with all six predictors from the correlation, and then dropping the least correlated one each time. To start the models are forecasted using the known values of the predictors from the test set. The best performing model will then be selected and tested with forecasted values for the predictors as well, to get at clear picture of a realistic forecasting situation.

The performance of each model is evaluated using both in- and out-of-sample metrics. The in-sample fit is evaluated using both AIC, that rewards goodness of fit and penalizes model complexity moderately, and BIC, that applies a stronger penalty for model complexity.

Tabel 3: Forecast Accuracy and Model Fit Comparison

Model	Predictors	RMSE	MAE	MAPE	ACF1	AIC	BIC
ARIMAX_model3	3	0.0300	0.0228	0.2130	0.4105	-421.1	-407.5
ARIMAX_model2	2	0.0380	0.0318	0.2962	0.4424	-421.1	-404.9
ARIMAX_model1	1	0.0461	0.0392	0.3660	0.6842	-417.9	-401.6
ARIMAX_model4	4	0.0466	0.0340	0.3186	0.5696	-425.1	-400.8
ARIMAX_model5	5	0.0471	0.0345	0.3232	0.5870	-423.6	-396.5
ARIMAX_model6	6	0.0491	0.0366	0.3429	0.5919	-422.1	-392.3
ARIMA_baseline	0	0.0673	0.0599	0.5589	0.8492	-404.8	-391.3

The out of sample forecast accuracy is measured using RMSE, MAE, MAPE and ACF1. The results shown in table 3 show the trade off between models complexity and forecasting accuracy. While the more complex models with 4, 5 and 6 predictors have a higher AIC, their complexity is penalised in the BIC, and are outperformed in accuracy by the simpler models. This could be a sign of over-fitting. In contrast the model with 3 predictors outperform the other models on all out of sample measures and still has very reasonable AIC and BIC values. Based on these results the ARIMAX_model3 is selected as the preferred model.



Figure 3: Residual plots and model report for ARIMAX_model3

The selected model is an ARIMA(1,1,0) error structure, meaning the first differenced log prices in Kbh S now depend on a single autoregressive part and the values of the exogenous variables. As seen in table 3 this model outperforms the baseline in both fit and accuracy. However this is based on the assumption, that the values of the predictors are known in the whole test period.

Tabel 4: Forecast Accuracy - Forecasted regressors

Model	RMSE	MAE	MAPE	ACF1
ARIMA_baseline	0.0673	0.0599	0.5589	0.8492
ARIMAX_model3	0.0722	0.0545	0.5083	0.7895

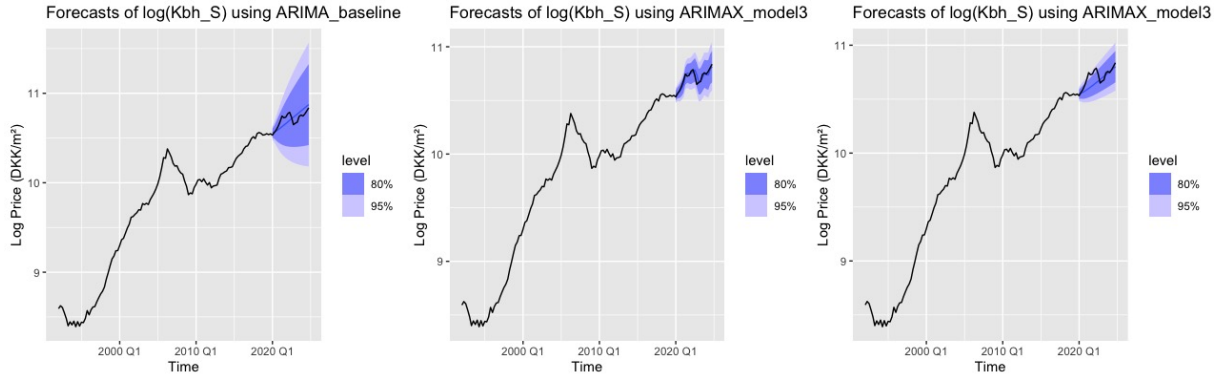


Figure 4: Forecasts of test set

This is not very realistic when forecasting further into the future, as the values for the regressors are only known one period in advance. To evaluate its forecasting abilities under realistic conditions, the model is tested using forecasted values for the exogenous variables.

The forecasts for the exogenous variables are generated using separate univariate ARIMA models for each predictor, before using these predicted values for the forecast in the ARIMAX_model3. The results in table 4 show that having to predict the values for the regressors significantly worsens the forecast accuracy. It no longer out performs the baseline model. The ARIMAX forecast still has a slightly lower MAE and MAPE than the baseline, but a higher RMSE, which suggests that the model performs well on average, but is prone to larger individual forecast error, which is a sign of reduced stability.

The plots of the forecasts in figure 4 show that the ARIMAX that uses the known values of the regressors follows the actual data nicely, as these values reflect the observed market movements directly. This feature is lost when we need to predict the values of the regressors as well, and it take the same shape as the baseline model. It is important to note, that the confidence bands on the forecast that uses these forecasted values is displayed very narrow compared to the baseline, this is because this plot does not take the uncertainty of the predicted regressors into consideration. In reality the confidence bands on this plot should be wider.

In conclusion adding the lagged values from other neighbourhood can be valuable, especially in the short term, when these may be known. However, for longer forecasts like we intend in this

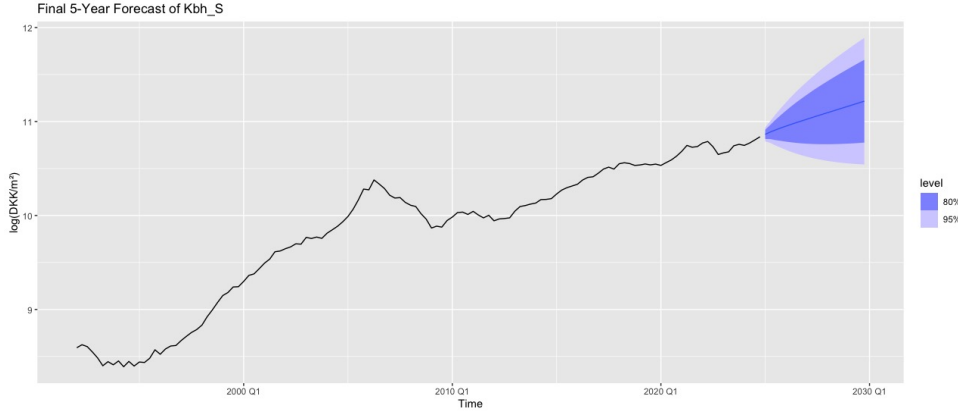


Figure 5: Final forecast using ARIMA

paper, the simplicity of the baseline ARIMA model would be preferred.

For the final forecast, the basic ARIMA model is re-estimated using the full dataset from 1992 Q1 to 2024 Q4. The new model selected is an ARIMA(2,1,0) compared to the test model ARIMA(1,1,2). The full model has an $AIC = -478$ and $BIC = -467$, which indicates an improved in-sample fit when the last five years are added. The resulting forecast in figure 5 look very reasonable with a continued upward trend and expanding confidence intervals as we forecast further into the future.

To see if an endogenous relationship between the areas could help, the VAR model is now considered.

5 VAR

To further investigate if other neighbourhoods around Kbh S can help improve the prediction, a VAR model is considered. Unlike the ARIMAX model, the VAR model allows for feedback between the neighbourhoods, which would more closely resemble the actual socioeconomic structures like spill overs or shared demand shocks.

For the VAR model all data needs to be stationary and we will therefore use the log differenced data from the neighbourhoods. Based on the correlations found in section 2.1 the neighbourhoods included in this VAR modelling will be Kbh Ø, Frb and Kbh N, as they showed the highest correlations with Kbh S. Including all neighbourhoods is avoided to prevent over-fitting and excessive model complexity, as the number of predicted parameters would inflate with the high dimensionality and reduce forecast stability without clear evidence that they could provide meaningful predictive power. Seven combinations of the three top predictor neighbourhoods and Kbh S will be tested, to see if including all three provide a better fit and prediction, than just one or two. The selection of the optimal lag length was done for each combination based on

Tabel 5: VAR Model Comparison – Forecast Accuracy and Fit

variables	p	residual_p	RMSE	MAE	AIC	HQ	SC	FPE
Kbh_S, Kbh_Ø	2	0.9276	0.0676	0.0602	-14.02	-13.92	-13.77	1e-06
Kbh_S, Frederiksberg, Kbh_N	2	0.2992	0.0716	0.0560	-20.53	-20.32	-20.01	0e+00
Kbh_S, Kbh_Ø, Frederiksberg	3	0.3964	0.0726	0.0561	-21.48	-21.17	-20.73	0e+00
Kbh_S, Frederiksberg	2	0.7005	0.0746	0.0565	-13.86	-13.75	-13.60	1e-06
Kbh_S, Kbh_Ø, Kbh_N	4	0.9064	0.0760	0.0685	-20.69	-20.30	-19.71	0e+00
Kbh_S, Kbh_N	4	0.5190	0.0781	0.0697	-13.10	-12.91	-12.64	2e-06
Kbh_S, Kbh_Ø, Frederiksberg, Kbh_N	4	0.8126	0.0805	0.0711	-28.31	-27.62	-26.60	0e+00

AIC using `vareselect()` in R. A limit of 5 lags was set, to limit the complexity. For all models the selected length was between 2 and 4. All models were then tested with a Portmanteau test using `serial.test()` to ensure the residuals are approximately white noise. Only models with $p > 0.05$ were retained. Lastly the remaining models are used to forecast 5 years ahead and compared to the actual data in the test set. The forecasts are then evaluated using RMSE and MAE.

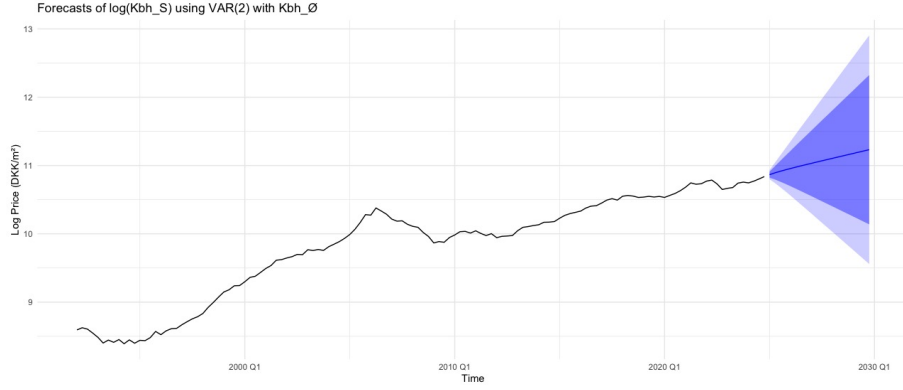
The evaluation results of the VAR models with different neighbourhood combinations are seen in table 5. The best performing model in terms of RMSE is the bivariate model containing Kbh S and Kbh Ø. It achieves the lowest RMSE of 0.0676 and a high p value for the Portmanteau test, indicating residuals that behave like white noise. The results also show that two of the trivariate models perform better in terms of MAE, but have slightly higher RMSE's this would mean that they perform better on average, but occasionally has some large errors. Interestingly the model containing all four neighbourhoods perform worst on both RMSE and MAE despite having the best in-sample fit. This could reflect over-fitting and model complexity that doesn't translate into improved forecasting.

To keep the focus on forecast accuracy, the RMSE will be prioritised, and the final selection of VAR model will therefore be the bivariate VAR(2) model containing only Kbh S and Kbh Ø. This model will then be used for the final forecast 5 years ahead based on all data available.

The VAR(2) model containing Kbh S and Kbh Ø is refitted to the full dataset, and tested with a Portmanteau test again. The test returns a p -value of 0.8605 which means we can't reject the null hypothesis and the residuals are accepted to be white noise. The forecast generated from the model is shown in figure 6 and looks reasonable with the continued increasing trend, and expanding confidence bands as we forecast further into the future.

6 Comparison

If we plot the two forecasts together, as seen in figure 7, we see that both follow approximately the same point-forecast line. This aligns with both models providing very similar RMSE's



Figur 6: Final forecast using VAR

(ARIMA:0.0673 VAR: 0.0676) when tested on the latest available data. This shows that the two models track the central trend similarly. However, the VAR model exhibits substantially wider confidence bands, indicating higher forecast uncertainty, likely due to the increased model complexity and uncertainty from multiple variables. In contrast the ARIMA model provides tighter confidence bands, more confident forecasts, making it more attractive. If we compare the VAR model to the selected ARIMAX model, that included Kbh Ø, Frb and Kbh N as exogenous regressors, we see a clear improvement when using a VAR setup, that considers the areas endogenous and allow for feedback, as it has less variation in the prediction errors. However, due to the simplicity and increased certainty, the ARIMA model would still be more attractive for long term forecasting.

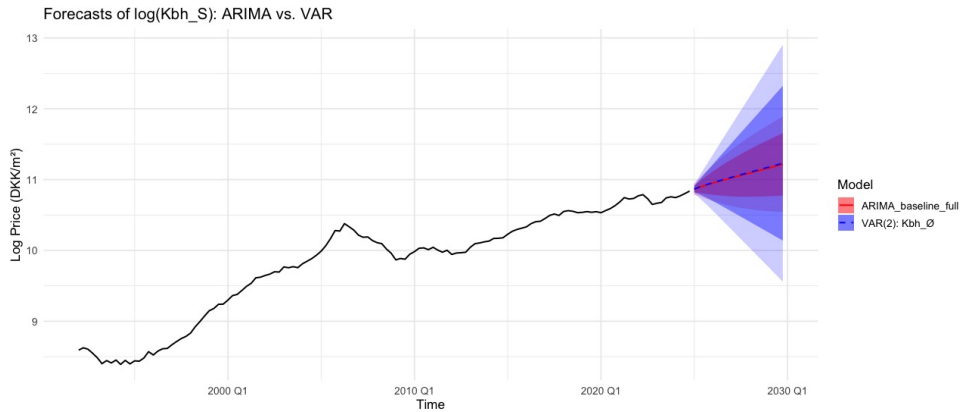


Figure 7: ARIMA and VAR forecasts

While the analysis and evaluation are done on logarithmic prices, it is often more intuitive to view forecasts in actual prices. When the predictions are transformed back to original scale, as shown in figure 8 the difference in forecast uncertainty becomes even more apparent. Especially the magnitude of the VAR models' confidence bands, making its uncertainty more visually

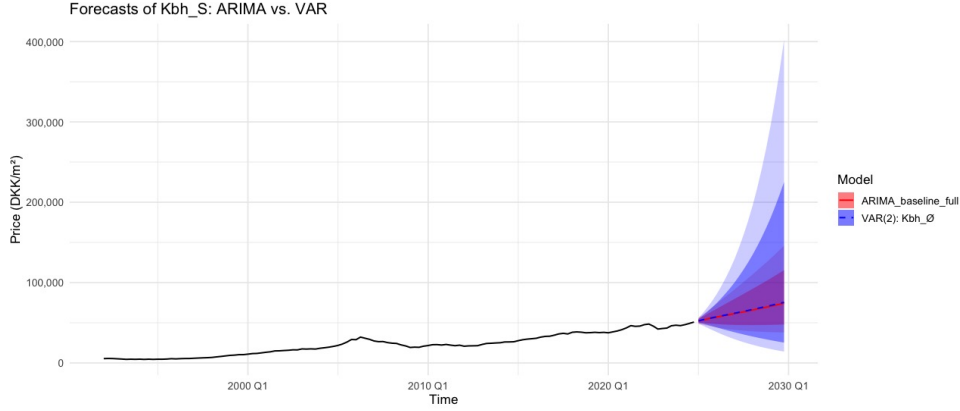


Figure 8: Final forecasts of the average square meter price in Kbh S

pronounced. This transformation reinforces the preference for the ARIMA model.

7 Conclusion

This paper analysed and forecasted the development in the average square meter prices for owner-occupied apartments sold in København S using ARIMA, ARIMAX, and VAR models. The baseline ARIMA model provided stable forecasts, while the ARIMAX model demonstrated that incorporating lagged prices from correlated neighbourhoods could improve short-term accuracy, but only when those prices are known. The VAR model allowed for mutual dynamics and yielded slightly better average performance, but at the cost of wider forecast uncertainty due to model complexity. When balancing performance, interpretability, and practical usability, the ARIMA model is the preferred approach for long-term forecasting of housing prices in Kbh S. Its simplicity, consistent accuracy, and narrower confidence intervals make it particularly suitable for stakeholders requiring robust projections with manageable uncertainty.

Another interesting addition could have been the inclusion of macroeconomic variables such as interest rates or new construction activity. These variables are often available or published in a greater advance, and may offer more practical value for long-term forecasting than recent price data from other neighbourhoods. Forecasting narratives in the news and media, for example, often focus on how national interest rate changes influence housing prices at a broad level. Such approaches provide valuable macroeconomic context and are well suited for understanding national or regional market trends. In contrast, the models applied in this paper focus on neighbourhood-level dynamics, offering a more localized perspective. Combining both approaches could provide a more comprehensive understanding of housing market developments.

A R Code

```
#####  
#           PA-EXAM           #  
#####  
  
# Load packages  
library(fpp3)  
library(xtable)  
library(tidyverse)  
library(AER)  
library(readxl)  
library(GGally)  
library(purrr)  
library(tsibble)  
library(gridExtra)  
library(urca)  
library(knitr)  
library(kableExtra)  
library(vars)  
library(dplyr)  
  
### Load data  
data<- read_excel('/Users/user/Library/KVMPRISER.xlsx', sheet = 1)  
head(data, 5)  
  
### Transform to tsibble  
data %>%  
  mutate(TID = yearquarter(TID)) %>%  
  as_tsibble(index=TID) -> tsKVMP  
  
head(tsKVMP, 5)  
  
# Log-transform all numeric variables  
log_prices <- tsKVMP %>%  
  mutate(across(where(is.numeric), log))  
  
#####  
## Preliminary Analysis ##  
  
### Plot raw data and ACF  
plot1a<- tsKVMP%>%  
  autoplot(Kbh_S) +  
  labs(title = "Square meter prices in Kbh S", y="Price in DKK", x="Time") +
```

```

scale_x_yearquarter(
  date_breaks = "2 years",
  expand      = c(0, 0)
)+
theme(
  plot.title = element_text(size = 16, face = "bold"),
  axis.title = element_text(size = 14),
  axis.text  = element_text(size = 10),
  legend.text = element_text(size = 12),
  legend.title = element_text(size = 13),
)

plot1b<- tsKVMP %>%
  ACF(Kbh_S)%>%
  autoplot() +
  theme(
    plot.title = element_text(size = 16, face = "bold"),
    axis.title = element_text(size = 14),
    axis.text  = element_text(size = 12),
    legend.text = element_text(size = 12),
    legend.title = element_text(size = 13),
  )

grid.arrange(plot1a,plot1b,nrow=2, heights= c(1.25,1))

### Plot Logdiff data and ACF
plot2a<- tsKVMP%>%
  autoplot(difference(log(Kbh_S))) +
  labs(title = "Logdiff Square meter prices in Kbh S", y="Log diff Prices", x="Time") +
  scale_x_yearquarter(
    date_breaks = "2 years",
    expand      = c(0, 0)
  )+
  theme(
    plot.title = element_text(size = 16, face = "bold"),
    axis.title = element_text(size = 14),
    axis.text  = element_text(size = 10),
    legend.text = element_text(size = 12),
    legend.title = element_text(size = 13),
  )

plot2b<- tsKVMP %>%
  ACF(difference(log(Kbh_S)))%>%
  autoplot() +
  theme(

```

```
plot.title = element_text(size = 16, face = "bold"),
axis.title = element_text(size = 14),
axis.text = element_text(size = 12),
legend.text = element_text(size = 12),
legend.title = element_text(size = 13),
)

grid.arrange(plot2a,plot2b,nrow=2, heights=c(1.25,1))

##### Stationarity for all areas #####
log_prices_diff <- log_prices %>%
  mutate(across(-TID, difference)) %>%
  filter(!is.na(Kbh_S)) # drop first row after differencing

areas <- colnames(log_prices_diff)[colnames(log_prices_diff) != "TID"]

stationarity_results <- data.frame(
  Area = character(),
  ADF_stat = numeric(),
  KPSS_stat = numeric(),
  ADF_crit_5pct = numeric(),
  KPSS_crit_5pct = numeric(),
  Stationary = character(),
  stringsAsFactors = FALSE
)

for (area in areas) {
  series <- log_prices_diff[[area]]

  # ADF test
  adf <- ur.df(series, type = "drift", selectlags = "AIC")
  adf_stat <- adf@teststat[1]
  adf_crit <- adf@cval[1,1]

  # KPSS test
  kpss <- ur.kpss(series, type = "mu")
  kpss_stat <- kpss@teststat
  kpss_crit <- kpss@cval[4]

  # Decision logic (require both tests to return valid values)
  if (adf_stat < adf_crit) {
    is_adf_stationary <- TRUE
  } else {
    is_adf_stationary <- FALSE
  }
}
```

```

}
if (kpss_stat < kpss_crit){
  is_kpss_stationary <- TRUE
} else {
  is_kpss_stationary <- FALSE
}
stationary_flag <- if (is_adf_stationary & is_kpss_stationary) "Yes" else "No"

stationarity_results <- rbind(
  stationarity_results,
  data.frame(
    Area = area,
    ADF_stat = round(adf_stat, 3),
    KPSS_stat = round(kpss_stat, 3),
    ADF_crit_1pct = round(adf_crit, 3),
    KPSS_crit_1pct = round(kpss_crit, 3),
    Stationary = stationary_flag
  )
)
}

print(stationarity_results)
stationarity_results%>%
  dplyr::select(-ADF_crit_1pct, -KPSS_crit_1pct)%>%
  t()%>%
kable(, format = "latex", digits = 3, booktabs = TRUE)

##### Cross - correlations #####
# Set up
lags <- -4:4
target <- "Kbh_S"
predictors <- setdiff(names(log_prices_diff), "TID")

# Define single-lag correlation function
ccf_single <- function(x, y, lag) {
  if (lag >= 0) cor(dplyr::lag(x, lag), y, use = "pairwise.complete.obs")
  else cor(dplyr::lead(x, -lag), y, use = "pairwise.complete.obs")
}

# Compute all correlations with Kbh_S as target
ccf_kbhs <- expand_grid(var_x = predictors, lag = lags) %>%
  mutate(cc = map2_dbl(var_x, lag, ~ ccf_single(log_prices_diff[[.x]], log_prices_diff[[target]], .y)))

# View top correlations

```

```

ccf_kbhs_use <- ccf_kbhs %>%
  filter(var_x != target, lag > 0)

predictive_lags2 <- ccf_kbhs %>%
  filter(var_x != target, lag > 0) %>%
  filter(abs(cc) > 0.3) %>%
  arrange(desc(abs(cc))) %>%
  slice_head(n = 6)

acf(log_prices_diff$Kbh_S, lag.max = 4, plot = FALSE)

knitr::kable(t(predictive_lags2), format = "latex", digits = 3, booktabs = TRUE)

#####
### ARIMA and ARIMAX ###
### log data #####
#####

### Split data
train_data <- log_prices %>% filter_index(. ~ "2019 Q4") # Up to and including 2019 Q4
test_data <- log_prices %>% filter_index("2020 Q1" ~ "2024 Q4") # 2020 Q1 to 2024 Q4

### Fit ARIMA and ARIMAX models on log-levels
models <- train_data %>%
  model(
    ARIMA_baseline = ARIMA(Kbh_S),
    ARIMAX_model6 = ARIMA(
      formula = Kbh_S ~ lag(Kbh_Ø, 1) + lag(Frederiksberg, 1) + lag(Kbh_N, 1) +
        lag(Frederiksberg_C, 2) + lag(Frederiksberg_C, 1) + lag(Kbh_K, 1) + pdq(0:5, 1, 0:5)
    ),
    ARIMAX_model5 = ARIMA(
      formula = Kbh_S ~ lag(Kbh_Ø, 1) + lag(Frederiksberg, 1) + lag(Kbh_N, 1) +
        lag(Frederiksberg_C, 2) + lag(Frederiksberg_C, 1) + pdq(0:5, 1, 0:5)
    ),
    ARIMAX_model4 = ARIMA(
      formula = Kbh_S ~ lag(Kbh_Ø, 1) + lag(Frederiksberg, 1) + lag(Kbh_N, 1) +
        lag(Frederiksberg_C, 2) + pdq(0:5, 1, 0:5)
    ),
    ARIMAX_model3 = ARIMA(
      formula = Kbh_S ~ lag(Kbh_Ø, 1) + lag(Frederiksberg, 1) + lag(Kbh_N, 1) +
        pdq(0:5, 1, 0:5)
    ),
    ARIMAX_model2 = ARIMA(
      formula = Kbh_S ~ lag(Kbh_Ø, 1) + lag(Frederiksberg, 1) + pdq(0:5, 1, 0:5)
    ),
    ARIMAX_model1 = ARIMA(

```

```

    formula = Kbh_S ~ lag(Kbh_0, 1) + pdq(0:5, 1, 0:5)
  )
)

### Baseline model report
models%>%
  dplyr::select(ARIMA_baseline)%>%
  report()

### Baseline model residual plot
models%>%
  dplyr::select(ARIMA_baseline) %>%
  gg_tsresiduals()

### All models glance
glance(models)

### Forecast on test set using ACUTAL FUTURE VALUES ###
forecast_values <- models %>%
  forecast(new_data = test_data)

### Forecast accuracy on test set
accuracy(forecast_values, test_data)

##### Comparison table#####
accuracy_table <- accuracy(forecast_values, test_data) %>%
  dplyr::select(.model, RMSE, MAE, MAPE, ACF1)

fit_table <- glance(models) %>%
  dplyr::select(.model, AIC, BIC)

### Join accuracy and fit summaries
model_summary <- left_join(accuracy_table, fit_table, by = ".model") %>%
  mutate(
    Num_Predictors = case_when(
      .model == "ARIMA_baseline" ~ 0,
      .model == "ARIMAX_model1" ~ 1,
      .model == "ARIMAX_model2" ~ 2,
      .model == "ARIMAX_model3" ~ 3,
      .model == "ARIMAX_model4" ~ 4,
      .model == "ARIMAX_model5" ~ 5,
      .model == "ARIMAX_model6" ~ 6,
      TRUE ~ NA_real_
    )
  ) %>%

```

```

dplyr::select(Model = .model, Num_Predictors, RMSE, MAE, MAPE, ACF1, AIC, BIC)%>%
mutate(
  AIC = round(AIC,1),
  BIC= round(BIC,1))%>%
arrange(RMSE)

### View table and print to latex
print(model_summary)
model_summary %>%
  arrange(RMSE) %>%
  kable(format = "latex", booktabs = TRUE, digits = 4,
        caption = "Forecast Accuracy and Model Fit Comparison")

### Model 3 report
models%>%
  dplyr::select(ARIMAX_model3)%>%
  report()

### Model 3 residual plots
models%>%
  dplyr::select(ARIMAX_model3) %>%
  gg_tsresiduals()

### Forecast plots - ACTUAL FUTURE VALUES ###
p1<-forecast_values %>%
  filter(.model=="ARIMA_baseline")%>%
  autoplot(log_prices, level = c(80, 95)) +
  labs(
    title = "Forecasts of log(Kbh_S) using ARIMA_baseline",
    y = "Log Price (DKK/m²)",
    x = "Time"
  )

p2<-forecast_values %>%
  filter(.model=="ARIMAX_model3")%>%
  autoplot(log_prices, level = c(80, 95)) +
  labs(
    title = "Forecasts of log(Kbh_S) using ARIMAX_model3 and known values of regressors",
    y = "Log Price (DKK/m²)",
    x = "Time"
  )

# Plot of all models using future values #
forecast_values %>%
  autoplot(log_prices, level = c(80, 95)) +

```

```

labs(
  title = "Forecasts of log(Kbh_S)",
  y = "Log Price (DKK/m2)",
  x = "Time"
)

### Forecast using FORECASTED FUTURE VALUES ###

### modelling predictors
model_Kbh0 <- train_data %>% model(Kbh_0 = ARIMA(Kbh_0))
model_Frb <- train_data %>% model(Frederiksberg = ARIMA(Frederiksberg))
model_KbhN <- train_data %>% model(Kbh_N = ARIMA(Kbh_N))

### forecasting predictors
fc_Kbh0 <- forecast(model_Kbh0, h = 20) %>%
  as_tibble() %>%
  dplyr::select(TID, Kbh_0 = .mean)

fc_Frb <- forecast(model_Frb, h = 20) %>%
  as_tibble() %>%
  dplyr::select(TID, Frederiksberg = .mean)

fc_KbhN <- forecast(model_KbhN, h = 20) %>%
  as_tibble() %>%
  dplyr::select(TID, Kbh_N = .mean)

## combine forecasted predictors
predictor_forecasts <- reduce(
  list(fc_Kbh0, fc_Frb, fc_KbhN),
  full_join,
  by = "TID"
)
predictor_forecasts <- predictor_forecasts %>%
  as_tsibble(index = TID)

### Forecast using ARIMAX_model3 with forecasted predictors
forecast_ex_ante <- models %>%
  dplyr::select(ARIMAX_model3, ARIMA_baseline) %>%
  forecast(new_data = predictor_forecasts)

### Plot forecast with forecasted future values
p3 <- forecast_ex_ante %>%
  filter(.model == "ARIMAX_model3") %>%
  autoplot(log_prices, level = c(80, 95)) +

```

```

labs(
  title = "Forecasts of log(Kbh_S) using ARIMAX_model3 and forecasted values of regressors",
  y = "Log Price (DKK/m2)",
  x = "Time"
)

### All test forecasts combined
grid.arrange(p1,p2,p3,ncol=3)

### Accuracy of model 3 and baseline using predicted values
accuracy(forecast_ex_ante, test_data) %>%
  dplyr::select(Model = .model, RMSE, MAE, MAPE, ACF1)%>%
  arrange(RMSE) %>%
  kable(format = "latex", booktabs = TRUE, digits = 4,
    caption = "Forecast Accuracy - Forecasted regressors")

#####
# Full Data ARIMA fit and forecast #

# Re-fit ARIMA baseline model to full dataset
full_model <- log_prices %>%
  model(ARIMA_baseline_full = ARIMA(Kbh_S))
report(full_model)

# Forecast 5 years (20 quarters) ahead
final_forecast <- full_model %>%
  forecast(h = 20)

# Plot
autoplot(final_forecast, log_prices) +
  labs(
    title = "Final 5-Year Forecast of Kbh_S ",
    y = "log(DKK/m2)",
    x = "Time"
  )

glance(full_model)

#####
##### END OF ARIMA #####

#####
### VAR selection #####

```

```

### LOG DIFF DATA #####
#####

### Setup
train_data <- log_prices_diff %>% filter(TID <= yearquarter("2019 Q4"))
test_data  <- log_prices_diff %>% filter(TID >= yearquarter("2020 Q1"))

train_matrix <- train_data %>%
  dplyr::select(-TID) %>%
  as.data.frame()

all_series <- colnames(train_matrix)
target <- "Kbh_S"
top_predictors <- c("Kbh_Ø", "Frederiksberg", "Kbh_N") # Based on correlation

### Create combinations: all 2- and 3-variable combos including Kbh_S
combos <- c(
  lapply(top_predictors, function(x) c(target, x)),
  combn(top_predictors, 2, simplify = FALSE) |> lapply(function(x) c(target, x)),
  list(c(target, top_predictors))
)

### Get last log(Kbh_S) from training set and actual log(Kbh_S) from test set
last_log <- log_prices %>% filter(TID == max(train_data$TID)) %>% pull(Kbh_S)
actual_log <- log_prices %>% filter(TID >= yearquarter("2020 Q1")) %>% slice(1:20) %>% pull(Kbh_S)

### Rprepare result matrix
results <- data.frame(
  variables = character(),
  p = integer(),
  residual_p = numeric(),
  rmse = numeric(),
  mae = numeric(),
  AIC = numeric(),
  HQ = numeric(),
  SC = numeric(),
  FPE = numeric(),
  stringsAsFactors = FALSE
)

#Loop through combos
for (vars in combos) {
  sub_matrix <- train_matrix %>%
    dplyr::select(all_of(vars)) %>%
    filter(if_all(everything(), is.finite))

```

```
try({
  cat("\n Trying model with:", paste(vars, collapse = ", "), "\n")

  var_selection <- VARselect(sub_matrix, lag.max = 5, type = "const")
  selected_p <- as.integer(var_selection$selection["AIC(n)"])
  cat(" Selected lag:", selected_p, "\n")

  if (is.na(selected_p) || !(selected_p %in% 1:ncol(var_selection$criteria))) {
    cat(" Invalid lag - skipping\n")
    next
  }

  var_model <- VAR(sub_matrix, p = selected_p, type = "const")
  cat(" VAR model fitted\n")

  resid_p <- serial.test(var_model, lags.pt = 16, type = "PT.asymptotic")$serial$p.value
  cat(" Residual p-value:", round(resid_p, 4), "\n")

  fc <- predict(var_model, n.ahead = 20)
  delta_fc <- fc$fcst$Kbh_S[, 1]
  log_fc <- cumsum(c(last_log, delta_fc))[-1]
  rmse <- sqrt(mean((actual_log - log_fc)^2))
  mae <- mean(abs(actual_log - log_fc))
  cat(" RMSE:", round(rmse, 4), "\n")

  ic <- var_selection$criteria
  aic <- ic["AIC(n)", selected_p]
  hq <- ic["HQ(n)", selected_p]
  sc <- ic["SC(n)", selected_p]
  fpe <- ic["FPE(n)", selected_p]
  cat(" Criteria extracted\n")

  results <- rbind(results, data.frame(
    variables = paste(vars, collapse = ", "),
    p = selected_p,
    residual_p = round(resid_p, 4),
    rmse = round(rmse, 4),
    mae = round(mae, 4),
    AIC = round(aic, 2),
    HQ = round(hq, 2),
    SC = round(sc, 2),
    FPE = round(fpe, 6)
  ))

  cat(" Model stored\n")
```

```
  }, silent = FALSE)
}

### Print sorted results
results_sorted <- results %>% arrange(rmse)
print(results_sorted)
results_sorted %>%
  arrange(rmse) %>%
  kable(format = "latex", booktabs = TRUE, digits = c(NA, 0, 4, 4, 4, 2, 2, 2, 8),
    caption = "VAR Model Comparison - Forecast Accuracy and Fit",
    row.names = FALSE)

##### SELECTED MODEL
### Identify best model's variable combination
best_vars <- strsplit(results_sorted$variables[1], ", ")[[1]]
best_p <- results_sorted$p[1]

### Subset the data accordingly
sub_matrix <- train_matrix %>%
  dplyr::select(all_of(best_vars)) %>%
  filter(if_all(everything(), is.finite))

### Fit the best VAR model and save it
best_model <- VAR(sub_matrix, p = best_p, type = "const")

### Extract residuals from the VAR model
residuals_var <- residuals(best_model)

### Plot residuals for Kbh_S
ts.plot(residuals_var[, "Kbh_S"], main = "Residuals of VAR model for Kbh_S", ylab = "Residuals")

### Autocorrelation plot
acf(residuals_var[, "Kbh_S"], main = "ACF of VAR Residuals for Kbh_S")

### Histogram
hist(residuals_var[, "Kbh_S"], breaks = 20, main = "Histogram of Residuals", xlab = "Residual", col = "gray")

##### PLOT VAR FORECAST ON TEST DATA
### Extract forecast and confidence intervals
fc <- predict(best_model, n.ahead = 20, ci = 0.95)

### Pull point forecast and standard errors
point_fc <- fc$fcst$Kbh_S[, 1]
lower_95 <- fc$fcst$Kbh_S[, 2]
```

```

upper_95 <- fc$fcst$Kbh_S[, 3]

# approximate 80% CI (1.28 * SE instead of 1.96)
se <- (upper_95 - point_fc) / 1.96
lower_80 <- point_fc - 1.28 * se
upper_80 <- point_fc + 1.28 * se

# Reconstruct log-level forecast from differences
log_fc <- cumsum(c(last_log, point_fc))[-1]
log_lower_95 <- cumsum(c(last_log, lower_95))[-1]
log_upper_95 <- cumsum(c(last_log, upper_95))[-1]
log_lower_80 <- cumsum(c(last_log, lower_80))[-1]
log_upper_80 <- cumsum(c(last_log, upper_80))[-1]

# get faorecast dates
forecast_dates <- test_data$TID[1:20]

# assemble
df_var_forecast <- tibble(
  TID = forecast_dates,
  value = log_fc,
  model = "VAR(2) with Kbh_0 on test",
  lower_80 = log_lower_80,
  upper_80 = log_upper_80,
  lower_95 = log_lower_95,
  upper_95 = log_upper_95
)

# Actual historical values
df_actual <- log_prices %>%
  dplyr::select(TID, value = Kbh_S) %>%
  filter(TID <= max(forecast_dates)) %>%
  as_tibble()

# Plot
ggplot() +
  geom_line(data = df_actual, aes(x = TID, y = value), color = "black") +
  geom_ribbon(data = df_var_forecast, aes(x = TID, ymin = lower_95, ymax = upper_95), fill = "blue", alpha = 0.2)
  geom_ribbon(data = df_var_forecast, aes(x = TID, ymin = lower_80, ymax = upper_80), fill = "blue", alpha = 0.4)
  geom_line(data = df_var_forecast, aes(x = TID, y = value), color = "blue") +
  labs(
    title = "Forecasts of log(Kbh_S) using VAR(2) with Kbh_0",
    y = "Log Price (DKK/m²)",
    x = "Time"
  ) +
  theme_minimal()

```

```
##### Plot test forecast with ARIMA
# Extract ARIMA_baseline forecast
fc_arima <- forecast_values %>%
  filter(.model == "ARIMA_baseline")

# Extract 80% and 95% intervals separately
fc_arima_80 <- fc_arima %>%
  hilo(80) %>%
  mutate(lower_80 = `80%`$lower,
         upper_80 = `80%`$upper) %>%
  dplyr::select(TID, .mean, lower_80, upper_80)

fc_arima_95 <- fc_arima %>%
  hilo(95) %>%
  mutate(lower_95 = `95%`$lower,
         upper_95 = `95%`$upper) %>%
  dplyr::select(TID, lower_95, upper_95)

# Combine both intervals
df_arima_full <- fc_arima_80 %>%
  left_join(fc_arima_95, by = "TID") %>%
  transmute(
    TID,
    model = "ARIMA_baseline",
    value = .mean,
    lower_80,
    upper_80,
    lower_95,
    upper_95
  )

# Combine forecast datasets
df_bands <- bind_rows(
  as_tibble(df_arima_full),
  as_tibble(df_var_forecast)
)

# Plot - test combined
ggplot() +
  # Confidence bands
  geom_ribbon(data = df_bands, aes(x = TID, ymin = lower_95, ymax = upper_95, fill = model), alpha = 0.2) +
  geom_ribbon(data = df_bands, aes(x = TID, ymin = lower_80, ymax = upper_80, fill = model), alpha = 0.4) +
  # Forecast lines
  geom_line(data = df_bands, aes(x = TID, y = value, color = model, linetype = model), size = 1) +
```

```

# Actual history
geom_line(data = df_actual, aes(x = TID, y = value), color = "black", size = 1) +
scale_color_manual(values = c("ARIMA_baseline" = "red", "VAR(2) with Kbh_0 on test" = "blue")) +
scale_fill_manual(values = c("ARIMA_baseline" = "red", "VAR(2) with Kbh_0 on test" = "blue")) +
scale_linetype_manual(values = c("ARIMA_baseline" = "solid", "VAR(2) with Kbh_0 on test" = "dashed")) +
labs(
  title = "Forecasts of log(Kbh_S): ARIMA vs. VAR",
  x = "Time",
  y = "Log Price (DKK/m2)",
  color = "Model",
  fill = "Model",
  linetype = "Model"
) +
theme_minimal(base_size = 8)

##### FINAL VAR PREDICTION #####

### adjust to full dataset
train_data <- log_prices_diff
train_matrix <- train_data %>%
  dplyr::select(-TID) %>%
  as.data.frame()
last_tid <- log_prices %>% filter(TID == max(train_data$TID)) %>% pull(TID)

# Subset the data accordingly
sub_matrix <- train_matrix %>%
  dplyr::select(all_of(best_vars)) %>%
  filter(if_all(everything(), is.finite))

### Fit the best VAR model and save it - final
best_model_final <- VAR(sub_matrix, p = best_p, type = "const")

### AIC
VARselect(sub_matrix, lag.max = 2, type = "const")$criteria[1,2]

# Portmanteau test - Final
serial.test(best_model_final, lags.pt = 16, type = "PT.asymptotic")

# Extract residuals from the VAR model - final
residuals_var <- residuals(best_model_final)

# Plot residuals for Kbh_S - final
ts.plot(residuals_var[, "Kbh_S"], main = "Residuals of VAR model for Kbh_S", ylab = "Residuals")

# Autocorrelation plot - final

```

```

acf(residuals_var[, "Kbh_S"], main = "ACF of VAR Residuals for Kbh_S")

# Histogram
hist(residuals_var[, "Kbh_S"], breaks = 20, main = "Histogram of Residuals", xlab = "Residual", col = "gray")

##### PLOT FINAL VAR
# Extract forecast and confidence intervals
fc <- predict(best_model_final, n.ahead = 20, ci = 0.95)

# Pull point forecast and standard errors
point_fc <- fc$fcst$Kbh_S[, 1]
lower_95 <- fc$fcst$Kbh_S[, 2]
upper_95 <- fc$fcst$Kbh_S[, 3]

# Approximate 80% CI (1.28 * SE instead of 1.96)
se <- (upper_95 - point_fc) / 1.96
lower_80 <- point_fc - 1.28 * se
upper_80 <- point_fc + 1.28 * se

# Reconstruct log-level forecast from differences
last_log <- log_prices %>% filter(TID == max(train_data$TID)) %>% pull(Kbh_S)
log_fc <- cumsum(c(last_log, point_fc))[-1]
log_lower_95 <- cumsum(c(last_log, lower_95))[-1]
log_upper_95 <- cumsum(c(last_log, upper_95))[-1]
log_lower_80 <- cumsum(c(last_log, lower_80))[-1]
log_upper_80 <- cumsum(c(last_log, upper_80))[-1]

### Find dates for forecast
forecast_dates <- seq(as.Date(as.yearqtr(as.character(last_tid)))) + months(3), by = "quarter", length.out = 20)
forecast_dates <- as.yearqtr(forecast_dates)

df_var_forecast <- tibble(
  TID = forecast_dates,
  model = "VAR(2): Kbh_Ø",
  value = log_fc,
  lower_80 = log_lower_80,
  upper_80 = log_upper_80,
  lower_95 = log_lower_95,
  upper_95 = log_upper_95
)

# 2. Actual historical values
df_actual <- log_prices %>%
  dplyr::select(TID, value = Kbh_S) %>%
  as_tibble()

```

```

# 3. Plot it all
ggplot() +
  geom_line(data = df_actual, aes(x = TID, y = value), color = "black") +
  geom_ribbon(data = df_var_forecast, aes(x = TID, ymin = lower_95, ymax = upper_95), fill = "blue", alpha = 0.2) +
  geom_ribbon(data = df_var_forecast, aes(x = TID, ymin = lower_80, ymax = upper_80), fill = "blue", alpha = 0.4) +
  geom_line(data = df_var_forecast, aes(x = TID, y = value), color = "blue") +
  labs(
    title = "Forecasts of log(Kbh_S) using VAR(2) with Kbh_0",
    y = "Log Price (DKK/m²)",
    x = "Time"
  ) +
  theme_minimal()

##### Plot with ARIMA
# Extract ARIMA_baseline final forecast
fc_arima <- final_forecast %>%
  filter(.model == "ARIMA_baseline_full")

# Extract 80% and 95% intervals separately
fc_arima_80 <- fc_arima %>%
  hilo(80) %>%
  mutate(lower_80 = `80%`$lower,
          upper_80 = `80%`$upper) %>%
  dplyr::select(TID, .mean, lower_80, upper_80)

fc_arima_95 <- fc_arima %>%
  hilo(95) %>%
  mutate(lower_95 = `95%`$lower,
          upper_95 = `95%`$upper) %>%
  dplyr::select(TID, lower_95, upper_95)

# Combine both intervals
df_arima_full <- fc_arima_80 %>%
  left_join(fc_arima_95, by = "TID") %>%
  transmute(
    TID,
    model = "ARIMA_baseline_full",
    value = .mean,
    lower_80,
    upper_80,
    lower_95,
    upper_95
  )

```

```

# Combine forecast datasets
df_var_forecast <- df_var_forecast %>%
  mutate(TID = tsibble::yearquarter(as.character(TID)))
df_bands <- bind_rows(
  as_tibble(df_arima_full),
  as_tibble(df_var_forecast)
)

# Plot
ggplot() +
  # Confidence bands
  geom_ribbon(data = df_bands, aes(x = TID, ymin = lower_95, ymax = upper_95, fill = model), alpha = 0.2) +
  geom_ribbon(data = df_bands, aes(x = TID, ymin = lower_80, ymax = upper_80, fill = model), alpha = 0.4) +
  # Forecast lines
  geom_line(data = df_bands, aes(x = TID, y = value, color = model, linetype = model), size = 0.7) +
  # Actual history
  geom_line(data = df_actual, aes(x = TID, y = value), color = "black", size = 0.7) +
  scale_color_manual(values = c("ARIMA_baseline_full" = "red", "VAR(2): Kbh_0" = "blue")) +
  scale_fill_manual(values = c("ARIMA_baseline_full" = "red", "VAR(2): Kbh_0" = "blue")) +
  scale_linetype_manual(values = c("ARIMA_baseline_full" = "solid", "VAR(2): Kbh_0" = "dashed")) +
  labs(
    title = "Forecasts of log(Kbh_S): ARIMA vs. VAR",
    x = "Time",
    y = "Log Price (DKK/m²)",
    color = "Model",
    fill = "Model",
    linetype = "Model"
  ) +
  theme_minimal(base_size = 13)

#####FINAL PLOT ARIMA AND VAR NOT LOG #####
#####

df_bands_exp <- df_bands %>%
  mutate(
    value = exp(value),
    lower_80 = exp(lower_80),
    upper_80 = exp(upper_80),
    lower_95 = exp(lower_95),
    upper_95 = exp(upper_95)
  )
df_actual_exp <- df_actual%>%
  mutate(
    value=exp(value)
  )

```

```

ggplot() +
  # Confidence bands
  geom_ribbon(data = df_bands_exp, aes(x = TID, ymin = lower_95, ymax = upper_95, fill = model), alpha = 0.2) +
  geom_ribbon(data = df_bands_exp, aes(x = TID, ymin = lower_80, ymax = upper_80, fill = model), alpha = 0.4) +
  # Forecast lines
  geom_line(data = df_bands_exp, aes(x = TID, y = value, color = model, linetype = model), size = 0.7) +
  # Actual history
  geom_line(data = df_actual_exp, aes(x = TID, y = value), color = "black", size = 0.7) +
  scale_color_manual(values = c("ARIMA_baseline_full" = "red", "VAR(2): Kbh_Ø" = "blue")) +
  scale_fill_manual(values = c("ARIMA_baseline_full" = "red", "VAR(2): Kbh_Ø" = "blue")) +
  scale_linetype_manual(values = c("ARIMA_baseline_full" = "solid", "VAR(2): Kbh_Ø" = "dashed")) +
  scale_y_continuous(labels = scales::comma)+
  labs(
    title = "Forecasts of Kbh_S: ARIMA vs. VAR",
    x = "Time",
    y = "Price (DKK/m²)",
    color = "Model",
    fill = "Model",
    linetype = "Model"
  ) +
  theme_minimal(base_size = 13)

```

```

#####
#####

```

B Data

TID	Kbh_K	Kbh_V	Frederiksberg_C	Frederiksberg	Kbh_Ø	Kbh_N	Kbh_S
1992Q1	7340	4667	6609	6504	5970	5834	5392
1992Q2	7751	4907	6487	6557	6305	5604	5568
1992Q3	6725	4792	6305	6452	6104	5527	5460
1992Q4	7134	4454	6270	6752	6280	5421	5153
1993Q1	6589	4325	6188	6184	5323	4938	4835
1993Q2	6531	4887	6322	6249	5618	5404	4451
1993Q3	6772	4885	6369	6444	5993	5252	4650
1993Q4	7301	4098	6582	6455	5953	4995	4497
1994Q1	7045	4313	6625	6297	6184	5104	4685
1994Q2	6958	4380	6626	6305	6147	5135	4406
1994Q3	6580	4490	6713	6492	5919	5426	4662
1994Q4	7175	4408	6623	6640	5918	5134	4438
1995Q1	6926	4335	6670	6676	6131	5313	4633
1995Q2	7595	4153	7145	6715	6214	5083	4610
1995Q3	7522	4774	7116	6715	6563	5573	4822

1995Q4	7895	4697	7520	6871	6522	5238	5275
1996Q1	8232	5075	7796	6985	6544	5837	5031
1996Q2	8306	5321	8132	7561	7149	6449	5320
1996Q3	8177	5491	8255	7882	7260	6179	5492
1996Q4	8358	5306	8199	7967	7292	6101	5520
1997Q1	9250	5625	9251	8435	8015	6896	5819
1997Q2	9896	6619	9537	8681	8348	7103	6082
1997Q3	9826	6775	9444	9009	8557	7662	6343
1997Q4	10564	7197	10020	9344	8976	7968	6543
1998Q1	11148	7641	10169	9661	9486	7839	6861
1998Q2	11588	8568	11604	10193	10180	8606	7494
1998Q3	12312	8956	11451	10694	10880	8857	8059
1998Q4	12526	9365	11142	10774	10921	9208	8729
1999Q1	13339	9924	12060	11563	11444	10348	9401
1999Q2	13653	10086	12830	12027	12046	10715	9706
1999Q3	15129	10876	13344	12850	12603	11512	10293
1999Q4	13084	10896	14011	12689	12727	11573	10322
2000Q1	14899	11458	13613	12788	12590	11702	10923
2000Q2	15248	13297	14140	13425	13631	12384	11642
2000Q3	16773	12727	14641	13659	14000	12277	11835
2000Q4	15965	13704	14394	14158	14073	13351	12520
2001Q1	17279	14366	15512	15133	14903	13985	13272
2001Q2	18314	14836	16271	15726	15672	14895	13847
2001Q3	18841	15713	16709	16078	16407	15723	14966
2001Q4	18232	15051	16401	16678	16637	15364	15097
2002Q1	19755	16325	17371	17217	16818	16295	15471
2002Q2	20131	17019	17508	17421	17561	16258	15764
2002Q3	20897	17306	18082	17617	17500	17338	16293
2002Q4	19956	17091	17859	17559	17408	16808	16234
2003Q1	22357	16981	18272	17501	18051	17326	17454
2003Q2	22806	17958	17766	17908	17750	17473	17247
2003Q3	21503	18547	18727	18669	18109	17992	17509
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2004Q1	22513	19542	19145	18826	19422	18399	18249
2004Q2	23563	20415	20136	19685	20326	19666	18919
2004Q3	26088	21556	21016	20639	21586	21163	19636
2004Q4	26488	22595	22584	21665	22950	21841	20601
2005Q1	27099	23766	24266	23265	24533	22975	21812
2005Q2	29562	26305	26621	25929	27627	24885	23554
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2005Q4	35044	30565	31136	30531	31337	29904	29174
2006Q1	36125	30930	32972	34114	33690	30568	28946
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2007Q1	36535	29453	32403	29413	30216	27966	27322

2007Q2	34714	28741	30795	29799	29378	27064	26520
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2008Q1	31281	26487	28423	26240	26136	24260	24574
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2021Q1	58667	55482	58110	51985	55575	50693	43616
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2023Q1	59355	54058	58445	48258	52945	48760	42878
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2023Q4	61461	60425	63650	52060	56055	52889	46962
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